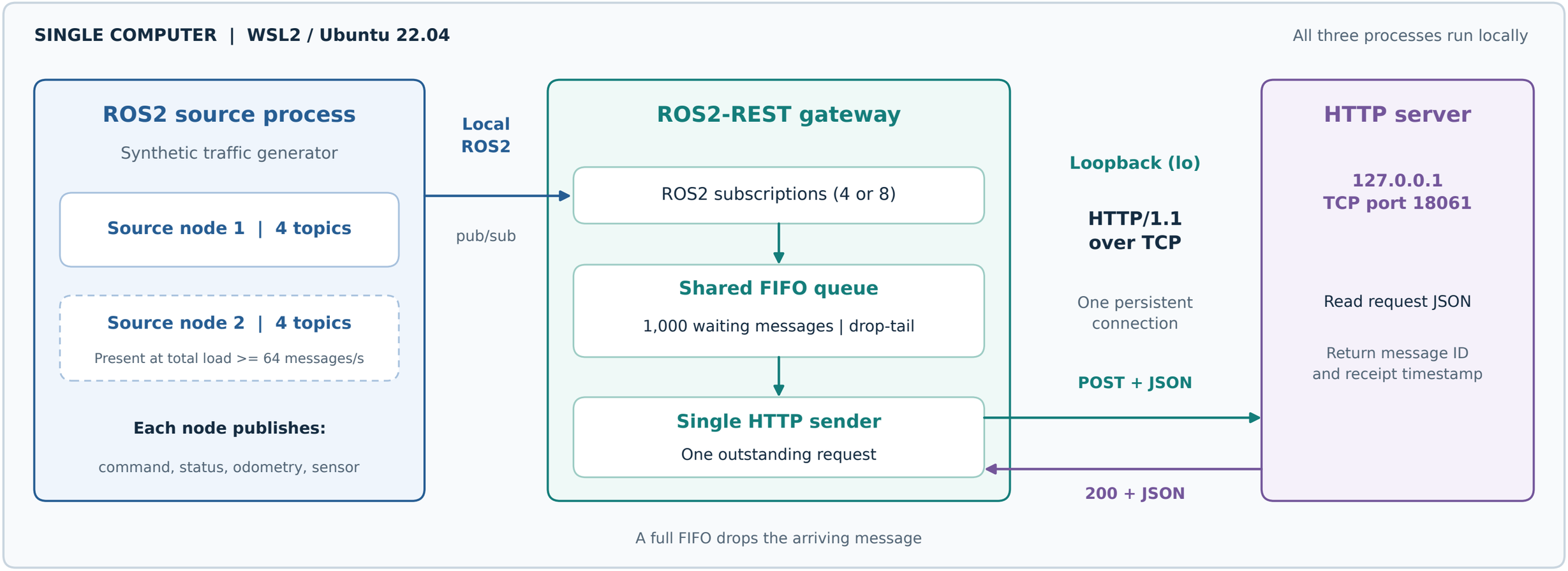
E1 historical loopback topology

Historical rate sweep: synthetic ROS2 traffic, one FIFO gateway, one HTTP server



LOWEST-LOAD TRAFFIC (ONE SOURCE)

command 2/s | status 5/s

odometry 20/s | sensor 5/s

Total: 32 messages/s

LOAD SCALING

Total: 32, 64, 128, 256, 512, 1,024, 2,048 messages/s

Above 32/s: two source nodes publishing progressively faster.

The HTTP client count stays at one throughout.

NETWORK CONDITIONS

No injected delay, loss or bandwidth limit.

The HTTP path stays inside this computer.